



TEMASEK JUNIOR COLLEGE  
PRELIMINARY EXAMINATION  
JC 2/ IP YEAR 6 2017

CANDIDATE  
NAME

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CENTRE  
NUMBER

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INDEX  
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## H2 BIOLOGY

Paper 1 Multiple Choice

**9744/01**

**Monday 18 September 2017**  
**1 hour**

Additional Materials: Multiple Choice Answer Sheet

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### READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice in **soft pencil** on the separate Multiple Choice Answer Sheet.

**Read the instructions on the Multiple Choice Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

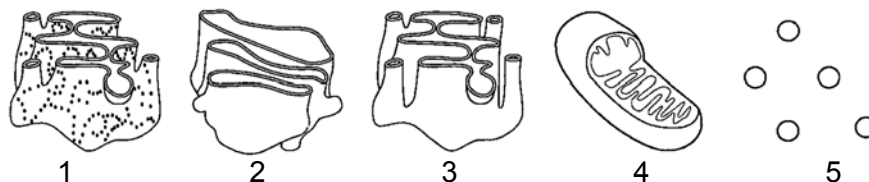
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

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This document consists of **17** printed pages and **1** blank page.

- 1 The diagram shows five different structures that can be observed in cells.



Which structures would be present in large quantities in a cell that is actively synthesising the following molecules?

|   | Extracellular glycolipids | Proteins   |
|---|---------------------------|------------|
| A | 1, 4, 5                   | 3, 4, 5    |
| B | 1, 3, 4, 5                | 1, 2, 4, 5 |
| C | 2, 3, 4, 5                | 1, 2, 4, 5 |
| D | 2, 3, 4, 5                | 1, 3, 4, 5 |

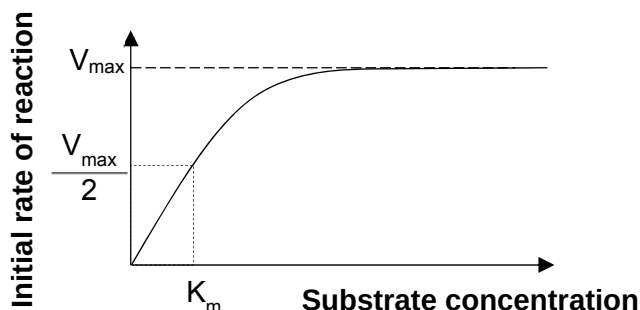
- 2 Keratin is a fibrous protein in skin, hair and nails. The features of one form of keratin are listed.

- 1 The peptide chain has mainly small amino acid residues.
- 2 Each peptide chain forms an  $\alpha$ -helix.
- 3 Two helices coil together.
- 4 Covalent bonds link adjacent helices.

Which features are the same in collagen molecule?

- A 1 and 2      B 1 and 4      C 2 and 3      D 3 and 4

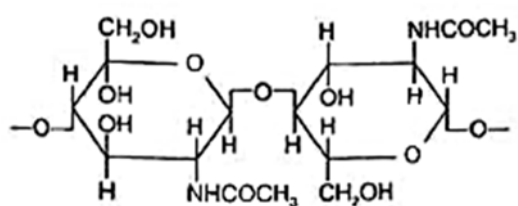
- 3 The value  $K_m$  is the substrate concentration at which the rate of an enzyme-catalysed reaction is half its maximum rate,  $\frac{V_{max}}{2}$ . The  $K_m$  was measured in the presence of a competitive inhibitor and a non-competitive inhibitor.



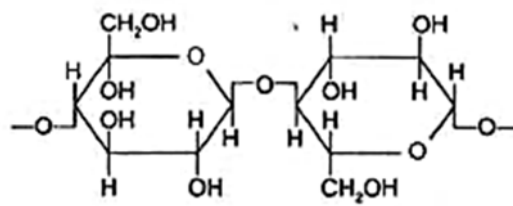
How will the value of  $K_m$  be affected in the presence of inhibitors?

|   | value of $K_m$ in presence of |                           |  |
|---|-------------------------------|---------------------------|--|
|   | competitive inhibitor         | non-competitive inhibitor |  |
| A | less                          | less                      |  |
| B | less                          | more                      |  |
| C | more                          | less                      |  |
| D | the same                      | more                      |  |

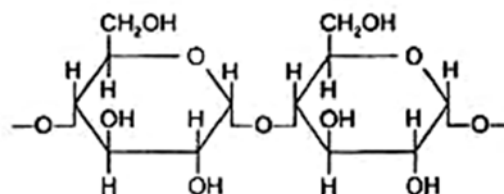
- 4 The diagrams show short sections of some common polysaccharides and modified polysaccharides.



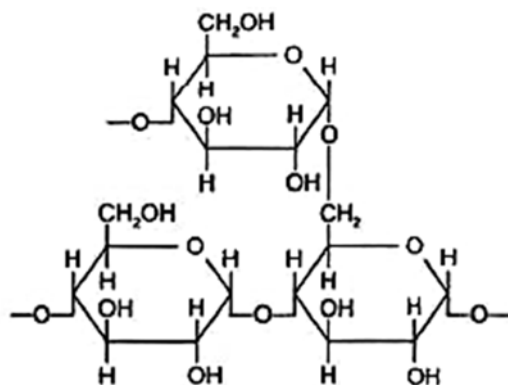
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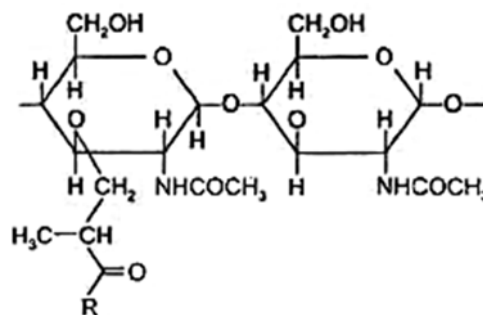
2



3



4



5

The polysaccharides can be described as below.

- polysaccharide F is composed of  $\beta$ -glucose monomers with 1,4 glycosidic bonds
- polysaccharide G is composed of  $\alpha$ -glucose monomers with 1,4 and 1,6 glycosidic bonds
- polysaccharide H is composed of N-acetylglucosamine and N-acetylmuramic acid monomers with  $\beta$ -1,4 glycosidic bonds
- polysaccharide J is composed of  $\alpha$ -glucose monomers with 1,4 glycosidic bonds
- polysaccharide K is composed of N-acetylglucosamine monomers with  $\beta$ -1,4 glycosidic bonds

Which shows the correct pairings of polysaccharide descriptions and diagrams?

|   | polysaccharide F | polysaccharide G | polysaccharide H | polysaccharide J | polysaccharide K |
|---|------------------|------------------|------------------|------------------|------------------|
| A | 2                | 4                | 5                | 3                | 1                |
| B | 2                | 5                | 4                | 1                | 3                |
| C | 3                | 4                | 1                | 2                | 5                |
| D | 3                | 5                | 4                | 1                | 2                |

- 5 Which of the following statement about membranes is correct?
- 1 All intracellular membranes in a eukaryotic cell have the same type of lipids and proteins.
  - 2 The outer and inner membranes of mitochondria have the same type of transport proteins.
  - 3 Carbohydrates form part of glycoproteins or glycolipids in the membranes.
  - 4 All plant cell membranes have cholesterol.

**A 3 only**

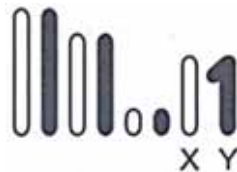
**B 1 and 4**

**C 2 and 3**

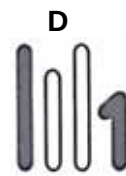
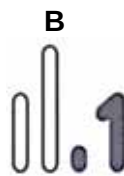
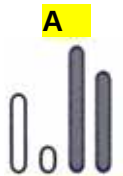
**D 1, 3 and 4**

- 6 No crossing over occurs during meiosis in male fruit flies of the species *Drosophila melanogaster*.

The diagram shows the four pairs of homologous chromosomes present in a testis cell of a male fly.



Which set of chromosomes in a gamete nucleus shows the genetic variation resulting from independent assortment?



- 7 What is the role of stem cells with regards to the function of adult tissues and organs?
- A** Stem cells are fully differentiated cells that reside under the surface of epithelial tissue, in position to take over the function of the tissue when the overlying cells become damaged or worn out.
  - B** Stem cells are totipotent cells that divide asymmetrically, giving rise to one daughter cell that remains a stem cell and one daughter cell that will differentiate to replace damaged and worn out cells in the adult tissue or organ.
  - C** Stem cells are embryonic cells that persist in the adult, and can give rise to all of the cell types in the body.
  - D** Stem cells are cells that have yet to express the genes and produce proteins characteristic of their differentiated state, but do so when needed for repair of tissues and organs.
- 8 A gene coding for an ion channel consists of 249 999 base pairs, which have 26 introns and 27 exons. During mRNA processing, a final transcript of 3570 bases is left.

How many additional amino acids would have been needed had the gene not contained introns?

**A 82 143**

**B 83 324**

**C 83 333**

**D 83 342**

- 9 Antibiotics are used to kill pathogens that infect people, without causing damage to human cells.

Different antibiotics work in different ways.

- Erythromycin binds to bacterial ribosomes.
- Nystatin binds to ergosterol which replaces cholesterol in pathogenic fungi.
- Rifampicin binds to bacterial RNA polymerase.
- Ciprofloxacin binds to DNA topoisomerase (enzyme that removes supercoiling of DNA).

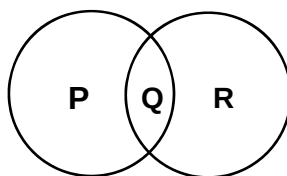
Which antibiotic directly inhibits the following process in pathogens?

|   | Membrane formation | DNA replication | Transcription | Translation   |
|---|--------------------|-----------------|---------------|---------------|
| A | rifampicin         | ciprofloxacin   | erythromycin  | nystatin      |
| B | rifampicin         | nystatin        | erythromycin  | ciprofloxacin |
| C | nystatin           | ciprofloxacin   | rifampicin    | erythromycin  |
| D | nystatin           | rifampicin      | ciprofloxacin | erythromycin  |

- 10 Which of the following statements about spliceosome and telomerase are correct?
- 1 Both function in the cytosol.
  - 2 The genes coding for spliceosome are found in the nucleolus whereas the genes coding for telomerase are found in other regions of the nucleus.
  - 3 Both are active in the inner cell mass of a blastocyst.
  - 4 The ribonucleotides of both ribonucleoproteins can form complementary base pairs.
  - 5 Spliceosomes are involved in both hydrolysis and condensation reaction, whereas telomerases are involved in condensation reaction only.
  - 6 After the extension of the telomeres by telomerase, the end replication problem partially caused by the spliceosome will not recur.

- A 1, 2, 5  
 B 1, 4, 6  
 C 2, 3, 6  
 D 3, 4, 5

- 11 Which row correctly describes the mode of control of gene expression in prokaryotes and eukaryotes?



- 1 Each gene is controlled by its own promoter.
- 2 Elongation continues after the release of the sigma factor.
- 3 The mRNA may contain several Shine-Dalgarno sequences.
- 4 Both processes of protein synthesis occur in the same location of the cell.
- 5 Attachment of RNA polymerase to promoter is achieved by interaction of transcription factors.
- 6 Binding of activators to enhancers increases the basal transcriptional activity.
- 7 Attachment of repressor proteins to specific regulatory sequence suppresses the basal transcriptional activity.
- 8 The level of activity of the newly synthesized protein is regulated by chemically modifying its structure.

|   | P                | Q    | R          |
|---|------------------|------|------------|
| A | 5, 6             | 7, 8 | 1, 2, 3, 4 |
| B | 2, 3, 5, 8       | 6, 7 | 1, 4       |
| C | 1, 6, 8          | 5, 7 | 2, 3, 4    |
| D | 1, 2, 5, 6, 7, 8 | 4    | 3          |

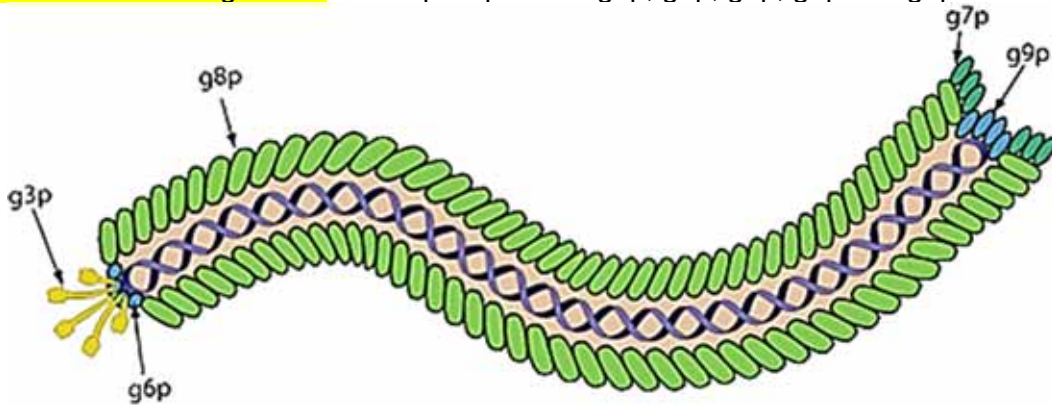
- 12 The following statements describe bacterial conjugation.

- 1 The F plasmid is made of single-stranded DNA.
- 2 When an F<sup>+</sup> donor gives its F plasmid to an F<sup>-</sup> recipient, both become F<sup>-</sup>
- 3 When an F<sup>+</sup> donor gives its F plasmid to an F<sup>-</sup> recipient, the donor becomes F<sup>-</sup>
- 4 When F<sup>+</sup> cells are mixed with F<sup>-</sup> cells, eventually all the cells will become F<sup>+</sup>.

Which of the following is correct?

- A 3 only
- B 4 only
- C 2 and 4
- D 1, 2 and 4

- 13 The diagram below shows the structure of an M13 bacteriophage. It consists of a **single-stranded circular DNA genome** and capsid proteins g3p, g6p, g7p, g8p and g9p.



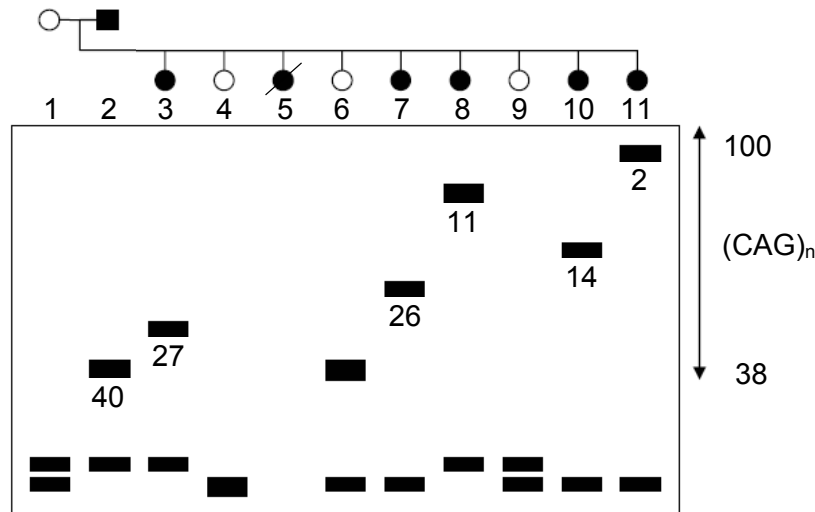
Based on your understanding of bacteriophages, which of the following statements is true of the M13 bacteriophage?

- 1 The base composition of its genome is such that the ratio of A:T is 1:1.
- 2 At least one of the capsid proteins is responsible for binding to a specific protein on the host cell.
- 3 Its genome is injected into the host cell after the phage attaches to the host cell.
- 4 It acquires its envelope from the cell membrane of its host cell.

- A 1 and 3  
**B 2 and 3**  
 C 1, 2 and 3  
 D 2, 3 and 4

- 14 The diagram below shows the results of electrophoresis of PCR fragments. Individuals with Huntington's disease have nucleotide sequence CAG that repeats from 36 to more than 120 times.

The male parent (individual 2) suffers from Huntington's disease when he was 40 years old. Six of his children (individuals 3, 5, 7, 8, 10, 11) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band containing the PCR fragments.



What conclusion can be drawn from the data above?

- A Individuals 4, 6, and 9 have not inherited the allele that causes Huntington's disease.  
 B Individuals 4, 6, and 9 will still develop Huntington's disease at some point in their lives, since the disease is inherited as a dominant trait.  
 C Individuals 4 and 9 do not have the trait, and will not get Huntington's disease, but individual 6 is likely to have the disease when she reaches her father's age of 40.  
 D Two of the three will develop the disease, since it is inherited as a dominant trait, but the data does not allow us to predict which two.
- 15 In most organisms, six different triplets of the DNA strand that is complementary to mRNA code for the amino acid serine: AGA, AGG, AGT, AGC, TCA and TCG.

In the yeast *Candida albicans*, a seventh DNA triplet, **GAC**, also codes for serine. In most organisms, this triplet codes for leucine. The diagram shows part of an mRNA molecule from *C. albicans*.

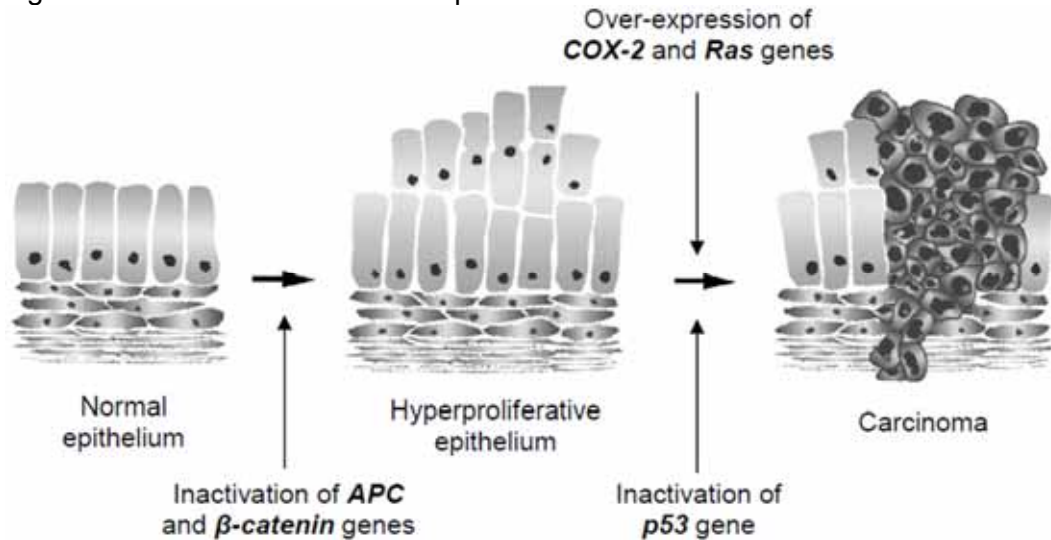
|              |     |     |            |     |     |     |     |
|--------------|-----|-----|------------|-----|-----|-----|-----|
|              | AGU | UCG | <b>CGG</b> | UCA | AGC | ACC | UGG |
| codon number | 11  | 12  | 13         | 14  | 15  | 16  | 17  |

Which mutation of the DNA that is complementary to this mRNA could result in *C. albicans* producing a polypeptide with a continuous sequence of five serines in it?

- A substituting a purine with a pyrimidine in the DNA coding for codon 13  
 B substituting a purine with a pyrimidine in the DNA coding for codon 16  
 C substituting a pyrimidine with a purine in the DNA coding for codon 13  
 D substituting a pyrimidine with a purine in the DNA coding for codon 16



- 16 The diagram below illustrates the development of colorectal cancer.



Which of these statements can be inferred from this multistep model of carcinogenesis?

- 1 Cells whose *APC* and  *$\beta$ -catenin* genes are inactivated have lost density dependent inhibition.
- 2 *APC* and  *$\beta$ -catenin* genes are most likely tumour suppressor genes.
- 3 High levels of *Ras* protein are produced only when both copies of *Ras* gene are mutated.
- 4 Two copies of normal *p53* alleles must be present to inhibit cell division.
- 5 Gain-of-function mutation in *COX-2* gene is one of the pre-requisites for the formation of carcinoma.

- A 1, 2 and 3  
**B 1, 2 and 5**  
 C 2, 3 and 4  
 D 2, 3 and 5

- 17 The speech defect known as stuttering may involve two genes, **G** and **N**. Most people are homozygous for the alleles **g** and **n** and are not stutterers.

However, recent research has shown that the presence of either of the mutant alleles **G** or **N** can cause stuttering in heterozygotes.

Using this information, which proportion of the children of a couple, the father with genotype **Ggnn** and the mother **ggNn**, are likely to be stutterers?

- A 3/16  
 B 8/16  
 C 9/16  
**D 12/16**

- 18 Which statement concerning chrysanthemum plants, of the genus *Dendranthema*, is a valid example of how the environment may affect the phenotype?
- A Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema*.
- B Identical genetic crosses performed between varieties of *Dendranthema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glasshouse.
- C The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parent plants.
- D The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5–15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0–25.0 cm) produce plants, when grown in natural day length, of a height varying between 55.0–71.0 cm.
- 19 The phenotype of the coat of a small mammal is controlled by two genes.
- The gene of hair colour has two alleles, **B** giving brown hair and **b** giving cinnamon hair.
  - The gene for hair shape has two alleles, **C** giving curly hair and **c** giving straight hair.

A number of cinnamon, straight-haired females were crossed with brown, curly-haired males who were homozygous for both genes. All the F<sub>1</sub> offspring showed the brown, curly-haired phenotype.

The F<sub>1</sub> offspring were back-crossed to animals with their mother's genotype. This resulted in 1000 offspring of four different genotypes.

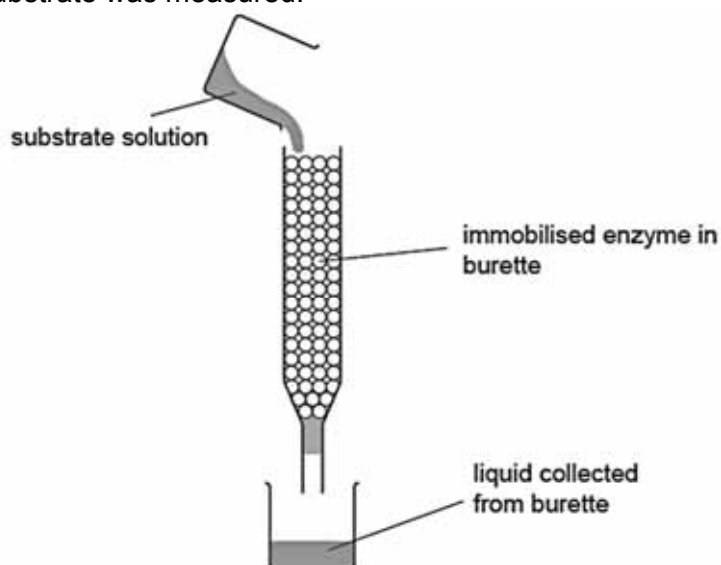
The table shows some of the results.

| genotype | number   |
|----------|----------|
| BbCc     | <b>R</b> |
| Bbcc     | <b>S</b> |
| <b>P</b> | 160      |
| <b>Q</b> | 340      |

Which row correctly identifies the missing genotypes (**P** and **Q**) and expected offspring numbers (**R** and **S**)?

|          | <b>P</b> | <b>Q</b> | <b>R</b> | <b>S</b> |
|----------|----------|----------|----------|----------|
| <b>A</b> | bbCc     | bbcc     | 160      | 340      |
| <b>B</b> | bbCc     | bbcc     | 340      | 160      |
| <b>C</b> | bbcc     | bbCc     | 160      | 340      |
| <b>D</b> | bbcc     | bbCc     | 340      | 160      |

- 20 A solution of a substrate was poured into a burette containing an enzyme immobilised onto alginate beads. The liquid passing through the burette was collected in a beaker and the concentration of substrate was measured.



The table below shows the results obtained by five students.

| The table below shows the results obtained by five students. |                                             |          |                         |          |
|--------------------------------------------------------------|---------------------------------------------|----------|-------------------------|----------|
| enzyme concentration / gdm <sup>-3</sup>                     | 0.2 / gdm <sup>-3</sup>                     |          | 0.4 / gdm <sup>-3</sup> |          |
|                                                              | substrate concentration / gdm <sup>-3</sup> |          |                         |          |
|                                                              | repeat 1                                    | repeat 2 | repeat 1                | repeat 2 |
| student A                                                    | 24                                          | 26       | 14                      | 13       |
| student B                                                    | 25                                          | 22       | 12                      | 12       |
| student C                                                    | 22                                          | 23       | 10                      | 13       |
| student D                                                    | 18                                          | 24       | 11                      | 12       |
| student E                                                    | 25                                          | 28       | 13                      | 18       |

A statistical test can be carried out to determine if the average substrate concentration collected for the two enzyme concentrations is significantly different.

Which of the following combination is correct?

|   | Degree of freedom | Conclusion                                                           |
|---|-------------------|----------------------------------------------------------------------|
| A | 3                 | If the p-value is greater than 0.5, the deviation is due to chance.  |
| B | 4                 | If the p-value is greater than 0.05, the deviation is due to chance. |
| C | 8                 | If the p-value is greater than 0.05, the deviation is due to chance. |
| D | 18                | If the p-value is greater than 0.5, the deviation is due to chance.  |

- 21** Bacteria in the genus *Wolbachia* infect many butterfly species. They are passed from one generation to the next in eggs, but not in sperm, and they selectively kill developing male embryos.

In Samoa in the 1960s, the proportion of male blue moon butterflies fell to less than 1% of the population. However, by 2006, the proportion of males was almost 50% of the population.

Resistance to *Wolbachia* is the result of the dominant allele of a suppressor gene.

Which statements correctly describe the evolution of resistance to *Wolbachia* in the blue moon butterfly population?

- 1 *Wolbachia* acts as a selective agent.
- 2 The selective killing of male embryos is an example of artificial selection.
- 3 When infected with *Wolbachia*, male embryos that are homozygous for the recessive allele of the suppressor gene die.
- 4 All male embryos that carry the dominant allele of the suppressor gene pass that allele to their offspring.
- 5 The frequency of the dominant allele of the suppressor gene rises in the butterfly population.

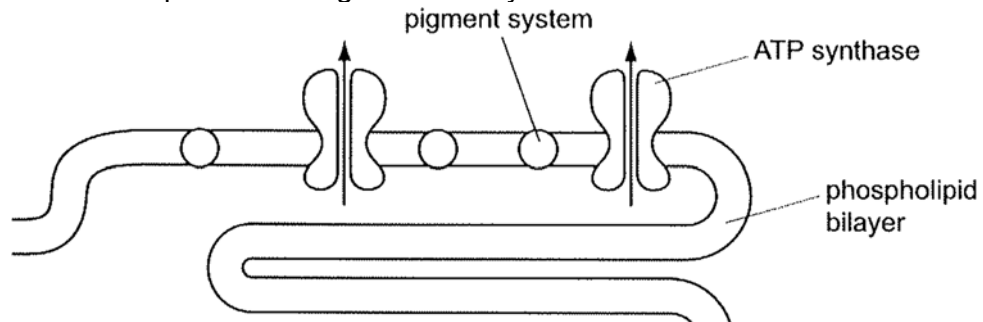
- A 1 and 4  
**B 1, 3 and 5**  
 C 2 and 3  
 D 2, 4 and 5

- 22** A biologist discovers two populations of wolf spiders whose members appear identical. Members of one population are found in the leaf litter deep within the woods. Members of the other population are found in the grass at the edge of the woods. The biologist decides to designate the members of the two populations as two separate species.

Which species concept has this biologist used and what is its limitation of this concept?

|          | Species concept                   | Limitation                                                                                                                                                |
|----------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | Morphological species concept     | Similarities in structures might have arisen due to convergent evolution.                                                                                 |
| <b>B</b> | Morphological species concept     | Cannot be used to group fossils or organisms which are completely asexual in their reproduction.                                                          |
| <b>C</b> | Ecological species concept        | Difficult to determine the magnitude of genetic variation required to distinguish between 2 putative species.                                             |
| <b>D</b> | <b>Ecological species concept</b> | <b>Difficult to determine what is considered as different niches, especially when organisms use resources from another niche during time of scarcity.</b> |

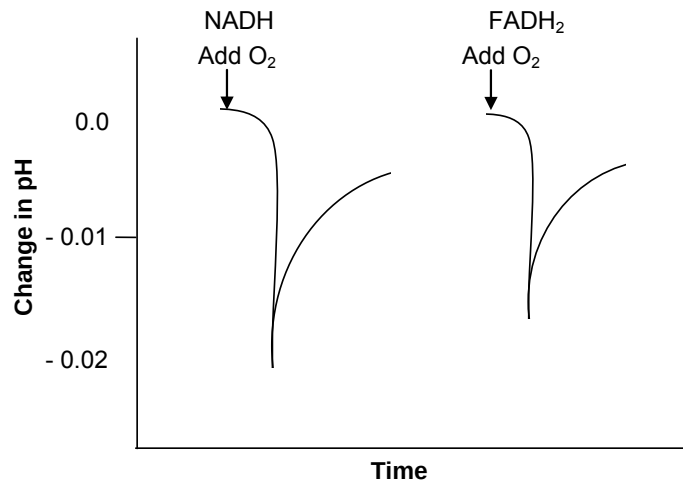
- 23 The diagram shows a small part of a thylakoid membrane. The arrows represent the movement of a particular reaction product through the ATP synthase.



From which chemical was this product derived from?

- A NADH      B NADPH      C Oxygen      **D Water**

- 24 Isolated mitochondria were incubated with NADH in one experiment and an equal amount of  $\text{FADH}_2$  in another set up. The mitochondria were initially deprived of oxygen. A known quantity of oxygen was then added and the pH of the intermembrane space was monitored. The result is shown in the graph.

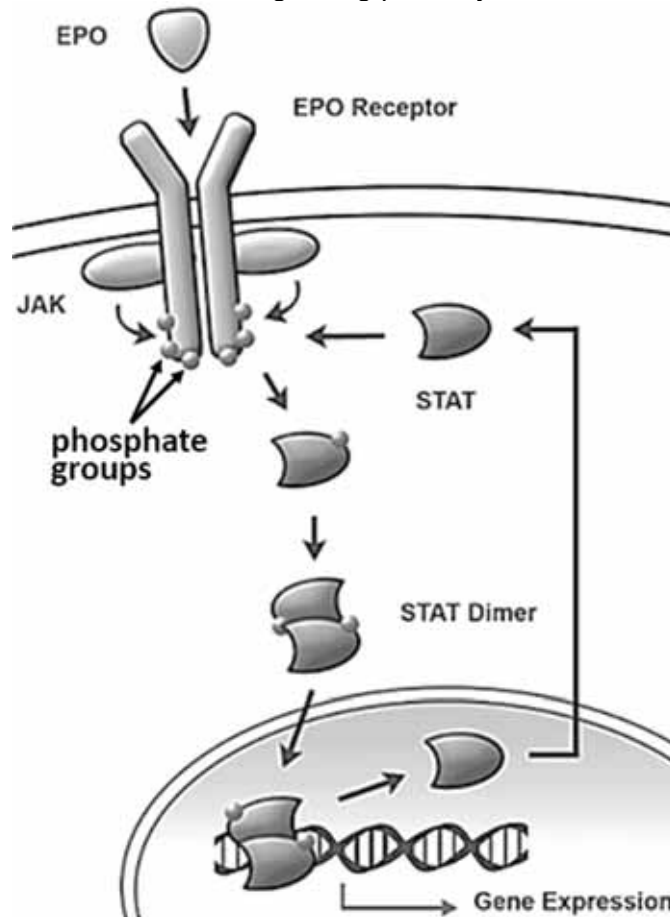


Which of the following can be concluded based on the results?

- 1 Upon the addition of oxygen, glycolysis and subsequently link reaction, Krebs cycle and oxidative phosphorylation occurred.
- 2 Electron transfer was initiated by the addition of oxygen.
- 3 The pH drop was greater with NADH than with  $\text{FADH}_2$ , which is consistent with the greater ATP yield that accompanies the oxidation of NADH.
- 4 The rapid decline in pH indicates that protons were pumped into the intermembrane space when oxygen was available.

- A 1 only  
 B 2 and 4 only  
**C 2, 3 and 4 only**  
 D All of the above

25 The diagram shows the JAK-STAT cell signalling pathway.

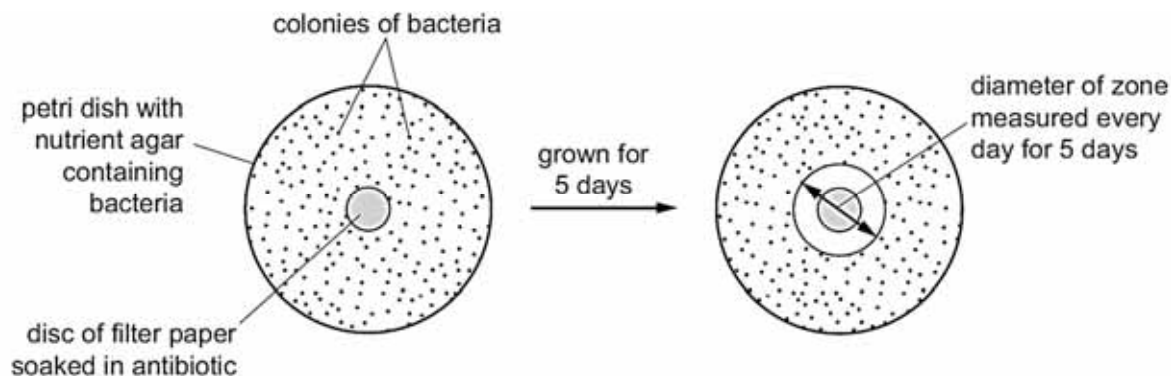


Which of the following statement is correct?

- 1 EPO is a type of steroid hormone.
- 2 Phosphorylation of STAT causes them to dimerize.
- 3 Gene expression is terminated when phosphatases remove phosphate groups from STAT dimers.
- 4 Signal amplification occurs as JAK phosphorylates multiple tyrosine residues on the EPO receptor.

- A 1 and 3 only  
**B 2 and 3 only**  
 C 2 and 4 only  
 D 2, 3 and 4 only

26 The diagram shows one way of testing the effect of an antibiotic on bacteria.



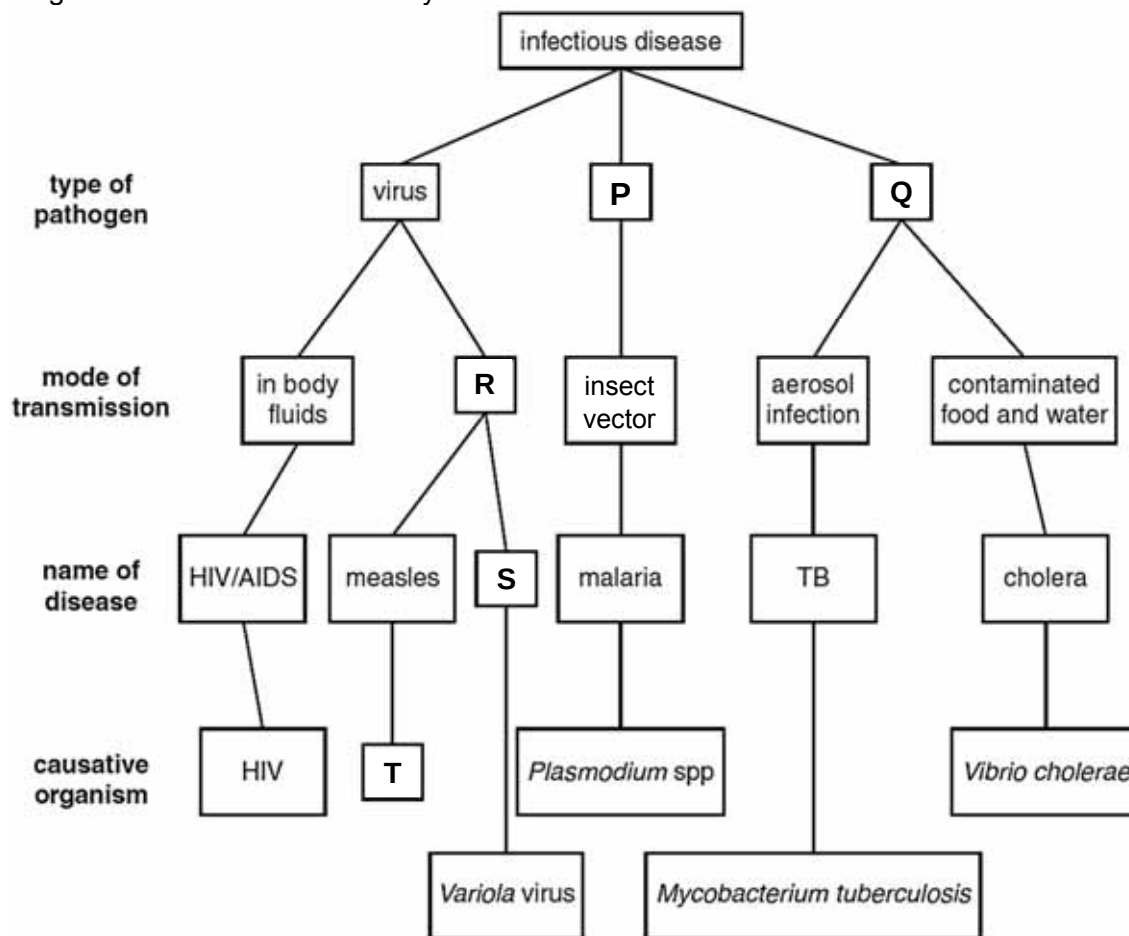
The table shows the results of testing five different types of bacteria. Zones of less than 13 mm show the presence of resistant bacteria.

| type of bacteria | diameter of zone / mm |       |       |       |       |
|------------------|-----------------------|-------|-------|-------|-------|
|                  | day 1                 | day 2 | day 3 | day 4 | day 5 |
| 1                | 24.10                 | 21.90 | 19.00 | 17.60 | 14.30 |
| 2                | 18.60                 | 15.40 | 12.20 | 9.00  | 0.00  |
| 3                | 17.90                 | 12.80 | 12.40 | 11.10 | 10.90 |
| 4                | 19.40                 | 15.30 | 13.20 | 8.10  | 0.00  |
| 5                | 22.00                 | 21.00 | 20.50 | 20.40 | 20.40 |

Which statement can be supported by this data?

- A All the types of bacteria become resistant to antibiotics over time.
- B Only types 2, 3 and 4 of the bacteria show resistance to the antibiotic.**
- C The antibiotic can be used to treat types 1 and 3 only.
- D Type 5 of the bacteria can never become resistant to the antibiotic.

27 The figure below shows a summary of some infectious diseases.



Which of the following combination is correct?

|   | P        | Q                | R                              | S                                |
|---|----------|------------------|--------------------------------|----------------------------------|
| A | Mosquito | Has cell wall    | Infection via aerosol droplets | Can be treated using antibiotics |
| B | Mosquito | Has mitochondria | Infection via sexual contact   | Can be prevented via vaccination |
| C | Protozoa | Has mitochondria | Infection via sexual contact   | Can be treated using antibiotics |
| D | Protozoa | Has cell wall    | Infection via aerosol droplets | Can be prevented via vaccination |



- 28** Rhesus (Rh) positive individuals have the Rh factor, an antigen present on the surface of their erythrocytes. Rh-negative individuals lack the Rh factor.

The Rh factor is of great medical importance especially for pregnant mothers who are Rh-negative and their foetus is Rh-positive. Their Rh-positive foetus may suffer from haemolytic disease, whereby the red blood cells are destroyed by the antibodies of the mother. The fetal blood and maternal blood are normally kept separate across the placenta. During delivery, a small amount of the baby's blood could come in contact with the mother's blood.

Which of the statement is correct?

- 1 The mother develops antibodies against the Rh factor after the first Rh-positive foetus is born.
- 2 The first Rh-positive foetus is less likely to suffer from haemolytic disease.
- 3 The subsequent Rh-positive foetus is likely to suffer from haemolytic disease, as the anti-Rh factor antibodies could cross the placenta and cause hemolysis of the Rh-positive fetal red blood cells.

- A** 1 only  
**B** 2 only  
**C** 2 and 3 only  
**D** All of the above

- 29** What is the impact of global warming on plants?

- A** In colder regions, a warmer climate may allow people to grow new crops.  
**B** Global warming beyond optimal growth temperatures encourages plant growth.  
**C** Temperate plants will shift to tropical regions.  
**D** Production of all crops are higher due to higher temperatures.

- 30** Which statement regarding adaptation of plants to global warming is false?

- A** Plants that develop longer roots, which allow for absorption of more water to counteract the loss of water through the leaves, are selected for.  
**B** Plants with fewer leaves, which reduce the effect of water loss through the leaves, are selected for.  
**C** When temperature increases, enzymatic activity increases, thus the rate of photosynthesis will increase.  
**D** When temperature increases, the leaves will respond by forming more stomata.